

RENA, Norway

WMO: 013890

Lat: 61.186N

Lon: 11.371E

Elev: 839

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-9.8	-18.8	2.0	-15.2	-13.1	2.8	-9.9	18.3	31.1	15.8	29.6	1.7	130	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	78.6	61.6	74.9	59.9	71.4	58.1	64.4	73.9	62.4	70.2	60.6	68.1	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	83.3	66.2	59.3	77.8	64.8	57.5	72.7	63.4	29.8	74.1	28.5	70.0	27.2	68.2	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.2	12.1	-19.2	83.0	6.5	3.7	-23.9	85.7	-27.7	87.8	-31.4	89.9	-36.1	92.6		
(5)			-21.2	66.3	6.7	3.1	-26.0	68.6	-29.9	70.4	-33.7	72.2	-38.5	74.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	15.8	18.1	27.9	36.9	46.4	54.2	58.9	56.4	48.3	37.9	27.3	18.2		
	DBStd	17.34	13.04	12.19	9.30	5.84	6.78	5.51	5.02	5.12	5.82	7.42	9.74	12.47		
	HDD50	5358	1062	893	684	393	152	20	2	8	99	380	680	985		
	HDD65	10122	1527	1313	1149	842	576	326	198	268	502	841	1130	1450		
	CDD50	724	0	0	0	1	41	146	279	207	46	4	0	0		
	CDD65	14	0	0	0	0	0	2	10	2	0	0	0	0		
	CDH74	381	0	0	0	0	36	85	202	58	0	0	0	0		
	CDH80	55	0	0	0	0	3	14	36	2	0	0	0	0		
	WSAvg	4.4	3.9	4.4	4.6	4.6	5.2	5.4	4.7	4.3	4.3	4.1	3.9	4.0		
	PrecAvg	29.1	2.1	1.5	1.1	1.7	2.6	3.0	3.6	3.9	2.8	2.6	2.5	1.9		
(15) Precipitation	PrecMax	38.9	4.2	3.7	3.3	3.8	5.9	6.0	8.4	7.6	5.6	6.3	8.0	3.5		
	PrecMin	21.6	0.4	0.3	0.0	0.1	0.0	0.2	0.8	1.0	1.1	0.1	0.6	0.2		
	PrecStd	4.2	1.0	0.7	0.8	1.0	1.5	1.3	1.5	1.5	1.1	1.5	1.6	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	43.1	52.7	64.1	78.8	82.7	84.6	79.2	69.9	58.9	50.0	44.4		
		MCWB	38.9	37.5	42.1	50.7	58.0	61.8	66.1	61.6	56.9	52.4	47.5	42.8		
	2%	DB	37.8	39.2	46.8	57.9	71.8	76.6	79.4	75.4	65.5	54.4	45.9	40.3		
		MCWB	36.1	36.2	38.8	46.4	54.9	59.7	63.8	60.5	55.9	50.1	44.2	38.7		
	5%	DB	34.7	36.2	43.1	53.2	67.0	71.8	75.5	71.8	61.8	51.8	42.5	37.2		
		MCWB	33.3	34.1	36.6	44.0	53.0	56.9	61.1	59.5	54.5	48.9	41.1	35.6		
	10%	DB	32.6	33.8	39.7	49.1	61.3	67.8	71.7	68.0	58.7	49.3	39.0	34.2		
		MCWB	31.8	32.2	35.0	41.5	49.9	55.3	59.2	58.4	53.2	46.8	38.1	33.2		
	0.4%	WB	39.6	39.1	42.8	52.2	60.0	63.9	68.0	66.1	60.8	54.7	48.3	43.2		
		MCDB	40.3	41.5	51.0	60.4	73.4	77.7	79.8	72.1	64.9	57.9	49.8	43.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.3	36.6	39.7	47.9	56.8	61.3	65.5	63.6	57.9	51.4	44.2	39.0		
		MCDB	37.4	38.5	45.3	55.3	68.2	72.2	76.1	70.0	62.4	53.3	45.3	39.9		
	5%	WB	33.6	34.4	37.7	45.3	54.3	59.2	63.3	61.7	55.9	49.6	41.2	35.7		
		MCDB	34.4	36.0	41.6	52.0	63.7	69.0	72.1	68.0	60.4	51.5	42.2	36.7		
	10%	WB	31.9	32.5	35.8	42.8	51.9	56.9	61.3	59.9	54.0	47.2	38.1	33.3		
		MCDB	32.6	33.8	39.2	48.5	60.4	64.9	68.8	66.2	57.7	48.9	38.8	34.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	14.9	18.5	19.2	21.9	21.0	20.4	19.6	17.8	12.8	9.8	11.3		
		MCDBR	11.0	14.6	22.1	28.2	34.1	30.7	29.2	27.4	23.9	16.4	12.0	12.8		
		MCWBR	10.4	12.0	14.5	17.4	17.9	14.9	14.1	14.5	15.0	12.0	11.3	11.8		
		MCDBR	10.4	13.2	19.1	24.6	28.5	25.0	24.1	19.9	20.1	12.6	11.7	12.1		
	5% WB	MCWBR	9.9	11.2	13.3	15.7	15.7	13.2	12.7	11.8	13.4	10.3	11.0	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.225	0.272	0.303	0.341	0.348	0.341	0.360	0.351	0.318	0.302	0.242	N/A		
	taud		2.146	2.260	2.268	2.303	2.430	2.481	2.442	2.461	2.535	2.485	2.231	N/A		
	Ebn at Noon		167	228	259	270	278	282	272	266	257	219	158	N/A		
	Edh at Noon		11	20	28	33	32	31	31	29	22	16	10	N/A		
(44) All-Sky Solar Radiation	RadAvg		89	300	792	1312	1505	1677	1519	1179	767	342	111	51		
	RadStd		15	59	74	135	147	153	193	105	93	60	26	5		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page